

Boyle's Law and Enzyme Kinetics

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PURPOSE

This investigation is broken down into two parts. The goal of **Part 1** is to investigate the relationship between pressure and volume of a gas at constant temperature. The goal of this investigation is to verify Boyle's Law. By varying the volume of a gas in a syringe and measuring the pressure with a gas sensor, this data can be used to confirm the inverse relationship showcased by Boyle's Law.

Part 2 investigates the kinetics of the enzyme catalase by measuring the rate of oxygen gas produced in the reaction with the hydrogen peroxide substrate. By recording the oxygen levels over time at different substrate concentrations, the initial reaction velocities can be analyzed using Michaelis-Menten kinetics. A Lineweaver-Burk plot can be constructed to extract kinetic parameters such as the maximum reaction velocity (V_{max}) and the Michaelis constant (K_m).

THEORY

Part 1: Boyle's Law

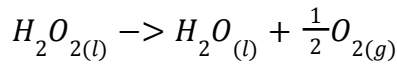
Boyle's Law describes the inverse relationship between the volume and pressure of a gas held at a constant temperature. It is mathematically represented as:

$$PV = \text{Constant}$$

In which P is the pressure, and V is the volume of a gas. Gas particles randomly move around their container; they occasionally slam into the walls of the container. This collision is referred to as pressure. If the volume of the container is decreased, the frequency at which these collisions happen will occur more often, which corresponds to an increase in pressure. This is assuming that the number of these particles and the temperature are not changing.

Part 2: Enzyme Kinetics

Catalase is an enzyme that accelerates the decomposition of hydrogen peroxide into water and oxygen:



Hydrogen peroxide decomposes at room temperature, but the rate is negligible over small time periods. The enzyme lowers the activation energy of the reaction without being consumed, binding the H_2O_2 to facilitate transformation into the products. Michaelis-Menten's equation can be used to describe the relationship between velocity (v) and substrate concentration (S):

$$v = \frac{V_{max}[S]}{(K_m + [S])}$$

V_{max} is the maximum reaction velocity (whenever the enzyme is fully saturated), and K_m is the Michaelis constant, which is equivalent to the substrate concentration at which $v = \frac{1}{2}V_{max}$. K_m is used to represent the binding affinity between the enzyme and the substrate. Since this is a hyperbolic relationship, to extract V_{max} and K_m , a Lineweaver-Burk plot is used, plotting $\frac{1}{v}$ on the y-axis and $\frac{1}{[S]}$ on the x-axis:

$$\frac{1}{v} = \left(\frac{K_m}{V_{max}}\right)\left(\frac{1}{[S]}\right) + \frac{1}{v_{max}}$$

This showcases the relationship where $\frac{1}{v_{max}}$ the y-intercept and the slope is $\frac{K_m}{v_{max}}$. The substrate concentration is prepared using serial dilutions from a 3% stock H_2O_2 solution $C_1V_1 = C_2V_2$.

EXPERIMENTAL

Part 1: Boyle's Law Theory

Equipment: Gas pressure sensor, Labquest Mini, Logger Pro Software, 60 mL syringe.

Procedure: Logger Pro was launched, and the gas pressure sensor was connected to the Labquest Mini, which was connected to the computer via USB.

The syringe was adjusted to 60mL of air before connecting to the valve. The gas pressure sensor adds a correction volume of 0.8mL to all syringe readings, so this needed to be taken into account.

The syringe was allowed to equilibrate to 60mL before recording the pressure. The syringe was compressed to each volume – 60mL, 50mL, and 40mL – for a one-minute period, then the pressures at each volume were recorded.

Part 2: Enzyme Kinetics Theory

Equipment: Oxygen Probe, Logger Pro software, 3% H_2O_2 stock solution, micropipette, and refrigerated catalase enzyme.

Logger Pro was configured to record oxygen levels for each trial for 60 seconds. Four different concentrations of hydrogen peroxide solutions were prepared during the experiment: 3.0%, 2.0%, 1.0%, and 0.5%. This was done by mixing the appropriate H_2O_2 volume with distilled water in the reaction container:

Table 1: Serial Dilutions

Target H_2O_2 (%)	Volume of 3% H_2O_2 (mL)	Volume of Water (mL)
3.0%	50.0	0
2.0%	33.3	16.7
1.0%	16.7	33.3
0.5%	8.3	46.7

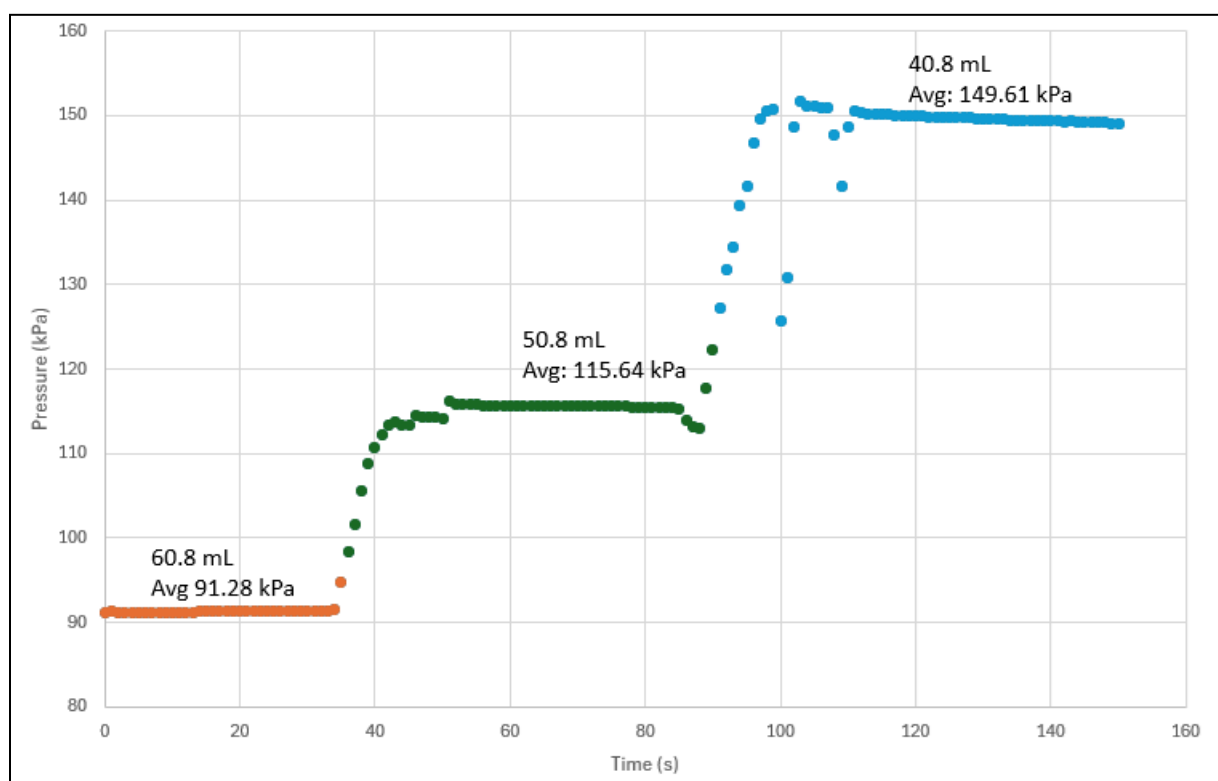
For each trial, 50 μ L of catalase was drawn into the pipette. Data collection on the Logger Pro software was started 10 seconds before the addition of the enzyme to establish a baseline. At the 10-second mark, the catalase was added, and the container was immediately sealed with the oxygen probe. The reaction was swirled for 2 seconds to ensure mixing. Oxygen levels were recorded for a 60-second duration.

DATA

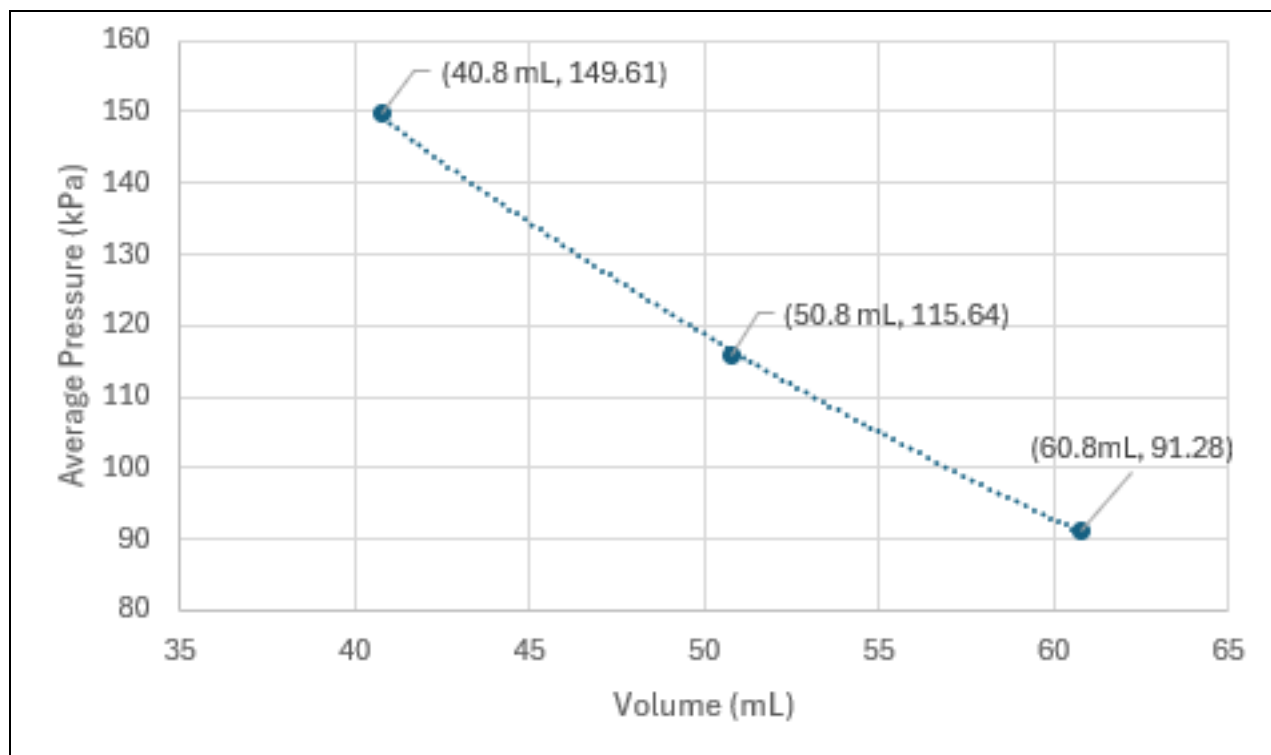
Part 1: Boyle's Law Data

Table 2: Data Summary

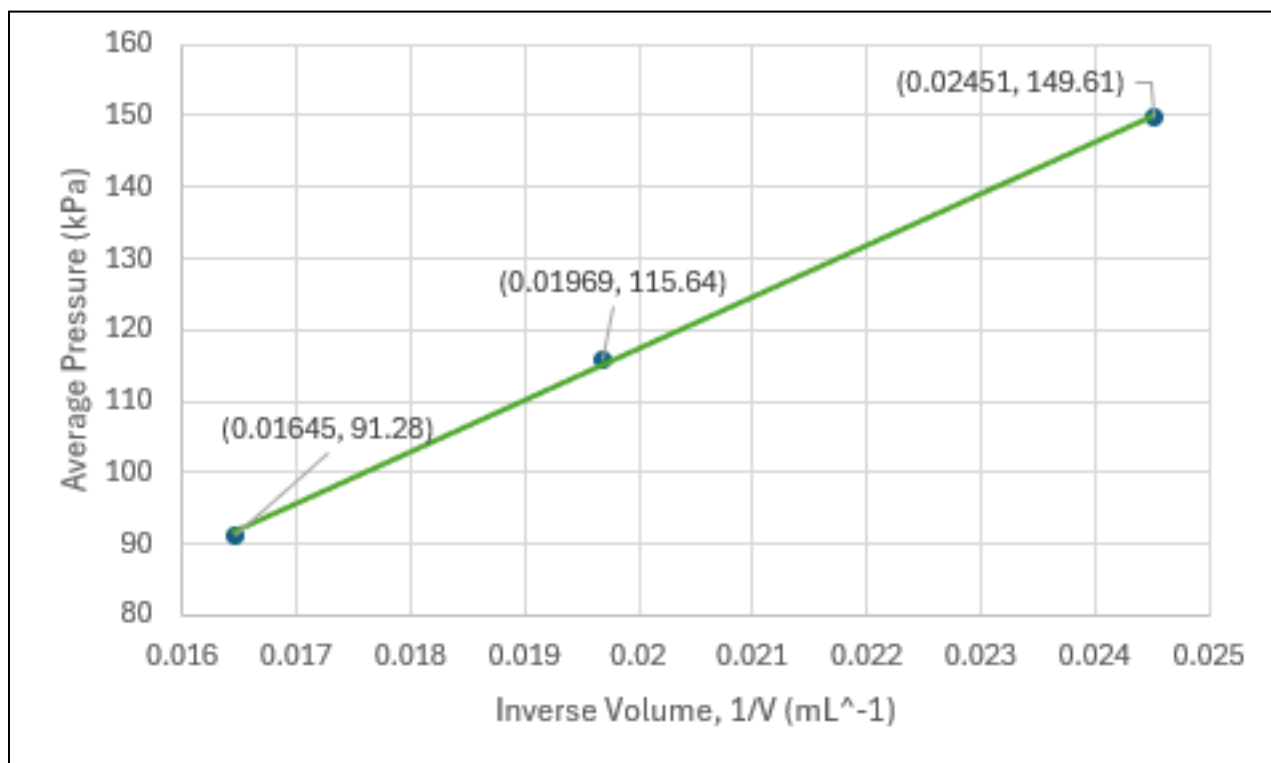
Volume (mL)	Avg Pressure (kPa)	$\frac{1}{v} (mL^{-1})$	PV (kPa*mL)
60.8	91.284	0.016447	5550.07
50.8	115.636	0.019685	5874.31
40.8	149.609	0.024510	6104.07



Graph 1: Pressure vs. Time



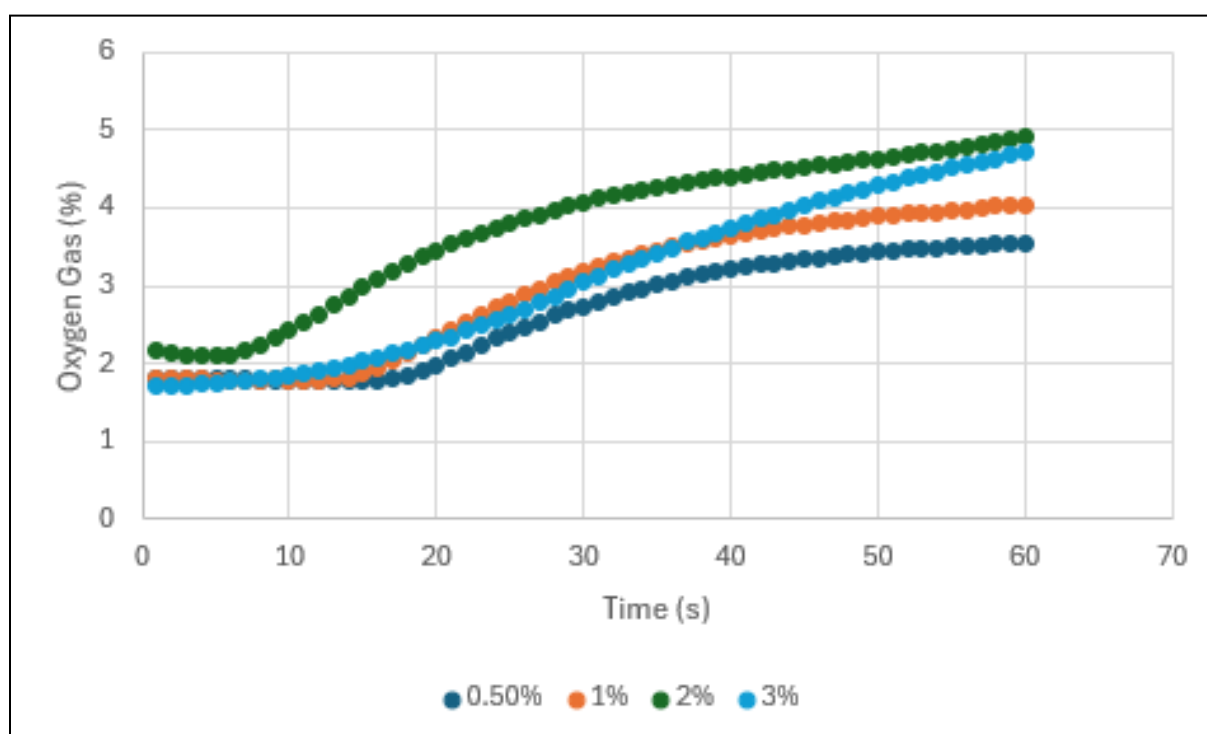
Graph 2: Average Pressure vs. Volume

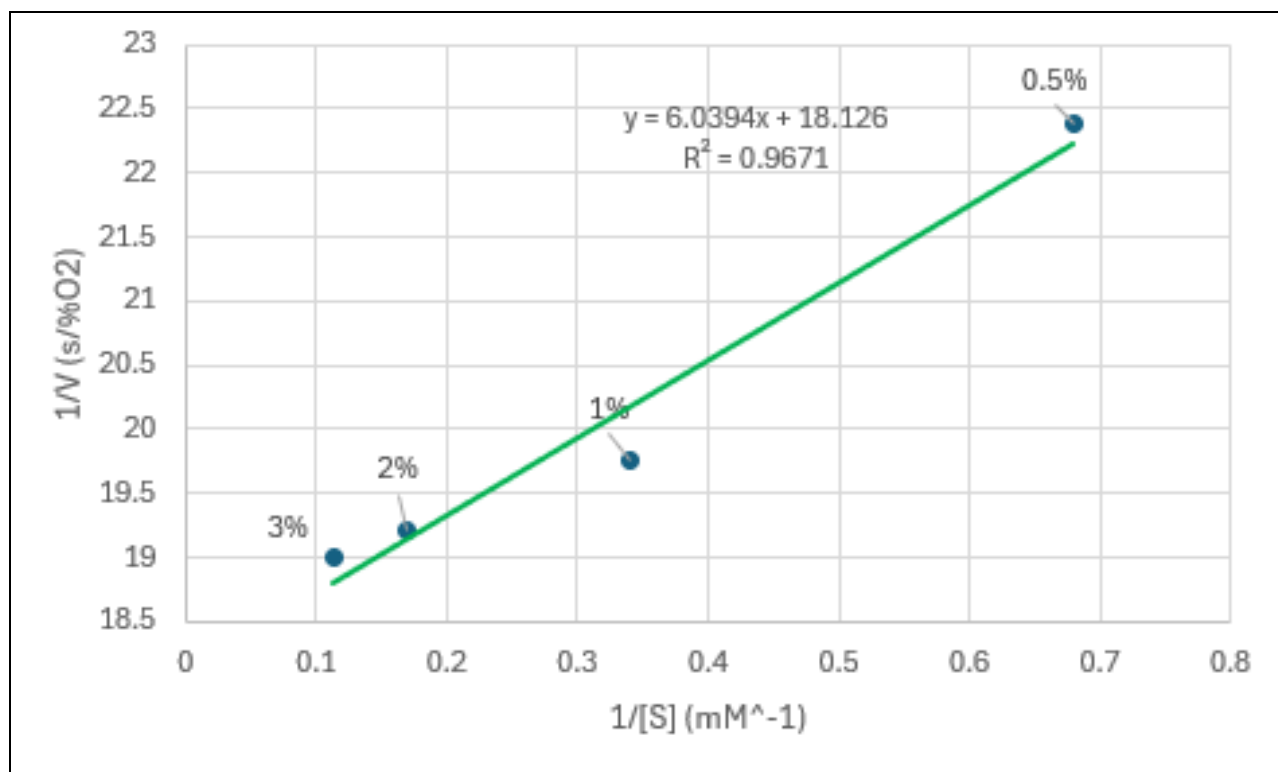


Graph 3: Average Pressure vs. $1/V$

Part 2: Boyle's Law**Table 3:** Kinetics Data Table

$[H_2O_2]\%$	$[H_2O_2]$ mM	V (% O_2 /s)	$1/[S]$ (mM^{-1})	$1/V$ (s/% O_2)
0.5%	1.4706	0.044655	0.6800	22.3938
1.0%	2.9412	0.050601	0.3400	19.7624
2.0%	5.8824	0.052045	0.1700	19.2142
3.0%	8.8235	0.052616	0.1133	19.0057

**Graph 4:** Oxygen Production Over Time for All H_2O_2 Concentrations



Lineweaver-Burk Plot - Catalase Kinetics

RESULTS

Part 1: Boyle's Law

Questions 1a & 1b - Data Table and Pressure Values

Table 2 summarizes the corrected volumes, average pressures, inverse volumes, and PV products recorded in the trial. Only one trial was completed due to time constraints; thus, the average pressure column reflects the single plateau value at each volume. The correction factor of 0.8 mL was applied to all syringe readings, yielding the actual volumes of 60.8, 50.8, and 40.8 mL.

As shown in **Graph 1** (Pressure vs. Time), the pressure at each volume was found by taking the average of the flat plateau region rather than a single value. This approach was chosen as the readings when slowly compressing the syringe do not accurately represent the stabilized values at the specific volumes. The plateau represents the thermally equilibrated state of the gas, which is a required condition of Boyle's Law. The corresponding pressures to volumes are 60.8 mL (91.28 kPa), 50.8 mL (115.64 kPa), and 40.8 (149.61 kPa).

Question 1c - Pressure vs. Volume Relationship

Graph 2 displays the average pressure against volume. As the volume decreased from 60.8 to 40.8, the pressure of the syringe increased, producing a hyperbolic curve. This is an inverse relationship, and as the volume increases, the pressure decreases, and vice versa. This is consistent with Boyle's Law.

Question 1d - Pressure vs. Inverse Volume Relationship

Graph 3 plots average pressure against inverse volume. This linearizes the data, creating a straight line. This confirms the predicted relationship between P and $1/V$. It demonstrates that the product PV remains approximately constant across the three volumes. The PV product values are 5550, 5874, and 6104; these are within an approximate range, given the experimental conditions: gas potentially leaking out of the syringe due to improper fitting, and only approximate volumes being reached due to the nature of a syringe.

Discussion Questions**Question 2 - Doubling the volume:**

Based on the conclusion drawn from the data collected in this experiment, doubling the volume of a gas sample would cause the pressure to decrease by half, assuming constant temperature equilibrates. This follows $PV = \text{Constant}$, for example, if 60.8 mL has a pressure of 91.28 kPa, doubling the volume to 121.6 mL would yield 45.64 kPa.

Question 3 - Effect of decreasing the volume to one-third:

If the volume of the gas is decreased to one-third of its original volume, the pressure can be expected to triple. Any operation that is done to P or V in the expression $PV = \text{constant}$, the equal and opposite operation has to be done to the other variable. Going from $V * \frac{1}{3}$ means that P needs to be multiplied by 3 to equal the same constant. Reducing 60.8 to 20.3 mL, the expected pressure would be 273.9 kPa.

Question 4 - Constants assumed from the ideal gas law:

The ideal gas law can be written as $PV = nRT$, where T is a constant temperature, R is the gas constant, and n is the number of moles of gas. Boyle's law assumes that n and T are held

constant, meaning that P and V are the two variables that are altered during this experiment. The moles of gas are constant due to the syringe being sealed, and the temperature is constant after the gas equilibrates after a volume change.

Part 2: Enzyme Kinetics

Questions 5 & 6 - Concentration Conversion and Dilution Calculations:

The four hydrogen peroxide solutions were prepared during the experiment by using serial dilution from a 3% stock solution. This was using the relationship $C_1V_1 = C_2V_2$, aiming for a final volume of 50 mL. Percent mass concentrations were converted to mM using the expression $C_{mM} = \left(\frac{C_{\%mass}}{34}\right) * 0.1 * 1000$, where 34 g/mol is the molar mass of hydrogen peroxide (H_2O_2), and the solution's density is approximated at 1 g/mL. These concentrations are reported in **Table 3**.

Question 7 - Oxygen Production

Graph 4 displays the Oxygen gas (%) vs. Time data for all four different concentrations of H_2O_2 . The data was trimmed so that only 10 seconds before the reaction and 50 seconds of the reaction are present in the datasets; this is the relevant window for the experiment. Higher concentrations produced steeper initial slopes as predicted. It is important to note that the enzyme added to the reaction (catalase) was older and may have been stored under sub-optimal conditions, leading to some degradation and potentially impacting the reaction rate. The lab protocol is written with 40 μ L added rather than the 50 μ L used in the experiment to compensate for this.

Questions 8 & 9 - Kinetics Data Table and Lineweaver-Burk Plot

The reaction velocities, substrate concentrations, and their reciprocals are reported in **Table 3**.

Velocities were calculated using $V = \frac{[O_2 \text{ at } t=30] - [O_2 \text{ at } t=10]}{20s}$, for 3% hydrogen peroxide, the data was trimmed to represent the appropriate timing in the experiment, then calculated using the same equation. **Graph 5**, the Lineweaver-Burk Plot, was created by plotting $1/V$ against $1/[S]$ for all four concentrations and fitting a trendline. The equation of the trendline was

$y = 6.0394x + 18.126$ ($R^2 = 0.9671$). From the y-intercept and slope, the V_{max} and K_m can be calculated:

$$V_{max} = \frac{1}{18.126} = 0.0552\% O_2/s$$

$$K_m = slope * V_{max} = 6.0394 * 0.0552 = 0.333mM$$

Question 10 - Interpretation and Comparing to Literature Values

The experiment resulted in a K_m value of 0.333mM, which is substantially lower than the literature value of approximately 93-94mM (Johnson, 2000). This error is expected as Johnson reports that relative errors of up to 400% are common in classroom demonstrations. The most likely source of error in the experiment is the storage and age of the enzyme, as a lower concentration of usable enzyme available will decrease the true rates of the enzyme. Another possible source of error could be the time between adding the enzyme and sealing the mixture, with the produced oxygen potentially escaping between sealing the reaction vessel.

REFERENCES

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